

BCEWithLogitsLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.loss.BCEWithLogitsLoss.html>

Description:

This loss combines a Sigmoid layer and the BCELoss in one single class. This version is more numerically stable than using a plain Sigmoid followed by a BCELoss as, by combining the operations into one layer, we take advantage of the log-sum-exp trick for numerical stability. The unreduced (i.e. with reduction set to 'none') loss can be described as:
$$-\sum_{i=1}^N [t_i \log(\sigma(x_i)) + (1 - t_i) \log(1 - \sigma(x_i))]$$
 where N is the batch size. If reduction is not 'none' (default 'mean'), then This is used for measuring the error of a reconstruction in for example an auto-encoder. Note that the targets $t[i]$ should be numbers between 0 and 1. It's possible to trade off recall and precision by adding weights to positive examples. In the case of multi-label classification the loss can be described as:

Function Signature

```
class torch.nn.modules.loss.BCEWithLogitsLoss(weight=None,  
size_average=None, reduce=None, reduction='mean',  
pos_weight=None)[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> target = torch.ones([10, 64], dtype=torch.float32) # 64
classes, batch size = 10
>>> output = torch.full([10, 64], 1.5) # A prediction (logit)
>>> pos_weight = torch.ones([64]) # All weights are equal to 1
>>> criterion =
torch.nn.BCEWithLogitsLoss(pos_weight=pos_weight)
>>> criterion(output, target) # -log(sigmoid(1.5))
tensor(0.20...)
```

```
>>> loss = nn.BCEWithLogitsLoss()
>>> input = torch.randn(3, requires_grad=True)
>>> target = torch.empty(3).random_(2)
>>> output = loss(input, target)
>>> output.backward()
```

Detailed Documentation

Documentation

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The unreduced (i.e. with reduction set to 'none') loss can be described as:

where NNN is the batch size. If reduction is not 'none' (default 'mean'), then

This is used for measuring the error of a reconstruction in for example an auto-encoder. Note that the targets $t[i]$ should be numbers between 0 and 1.

It's possible to trade off recall and precision by adding weights to positive examples. In the case of multi-label classification the loss can be described as:

where ccc is the class number ($c > 1$ for multi-label binary classification, $c = 1$ for single-label binary classification), nnn is the number of the sample in the batch and pcp_cpc is the weight of the positive answer for the class ccc .

$pc > 1$ increases the recall, $pc < 1$ increases the precision.

For example, if a dataset contains 100 positive and 300 negative examples of a single class, then pos_weight for the class should be equal to $\frac{300}{100} = 3$. The loss would act as if the dataset contains $3 \times 100 = 300$ positive examples.

In the above example, the pos_weight tensor's elements correspond to the 64 distinct classes in a multi-label binary classification scenario. Each element in pos_weight is designed to adjust the loss function based on the imbalance between negative and positive samples for the respective class. This approach is useful in datasets with varying levels of class imbalance, ensuring that the loss calculation accurately accounts for the distribution in each class.

weight (Tensor, optional) – a manual rescaling weight given to the loss of each batch element. If given, has to be a Tensor of size $nbatch$.

size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field $size_average$ is set to False, the losses are

instead summed for each minibatch. Ignored when reduce is False. Default: True

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

pos_weight (Tensor, optional) – a weight of positive examples to be broadcasted with target. Must be a tensor with equal size along the class dimension to the number of classes. Pay close attention to PyTorch's broadcasting semantics in order to achieve the desired operations. For a target of size [B, C, H, W] (where B is batch size) pos_weight of size [B, C, H, W] will apply different pos_weights to each element of the batch or [C, H, W] the same pos_weights across the batch. To apply the same positive weight along all spatial dimensions for a 2D multi-class target [C, H, W] use: [C, 1, 1]. Default: None

Input: $(*)^{(*)} (*)$, where $***$ means any number of dimensions.

Target: $(*)^{(*)} (*)$, same shape as the input.

Output: scalar. If reduction is 'none', then $(*)^{(*)} (*)$, same shape as input.

Runs the forward pass.

